PG No. 31 $T_{\rm d}$ $\bar{4}3m$ [cubic]

* generator : $2_{001},\ 2_{010},\ 3^{+}_{111},\ \mathrm{m}_{1-10}$
* conjugacy class

$[1]:$ 1
$[2_{001}]:$ $2_{001},\ 2_{100},\ 2_{010}$
$[3^{+}_{111}]:$ $3^{+}_{111},\ 3^{+}_{1-1-1},\ 3^{+}_{-11-1},\ 3^{+}_{-1-11},\ 3^{-}_{111},\ 3^{-}_{1-1-1},\ 3^{-}_{-11-1},\ 3^{-}_{-1-11}$
$[\mathrm{m}_{110}]:$ $\mathrm{m}_{110},\ \mathrm{m}_{101},\ \mathrm{m}_{011},\ \mathrm{m}_{1-10},\ \mathrm{m}_{-101},\ \mathrm{m}_{01-1}$
$[-4^{+}_{001}]:$ $-4^{+}_{001},\ -4^{+}_{100},\ -4^{+}_{010},\ -4^{-}_{001},\ -4^{-}_{100},\ -4^{-}_{010}$

* symmetry operation

Table 1: Symmetry operations for 3d polar vector.

No.	tag	matrix (polar)	det
1	1	$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$	1
2	2_{001}	$\begin{bmatrix}-1&0&0\\0&-1&0\\0&0&1\end{bmatrix}$	1
3	2_{010}	$\begin{bmatrix}-1&0&0\\0&1&0\\0&0&-1\end{bmatrix}$	1
4	2_{100}	$\begin{bmatrix}1&0&0\\0&-1&0\\0&0&-1\end{bmatrix}$	1
5	3^{+}_{111}	$\begin{bmatrix}0&0&1\\1&0&0\\0&1&0\end{bmatrix}$	1
6	3^{+}_{-11-1}	$\begin{bmatrix}0&0&1\\-1&0&0\\0&-1&0\end{bmatrix}$	1
7	3^{+}_{1-1-1}	$\begin{bmatrix}0&0&-1\\-1&0&0\\0&1&0\end{bmatrix}$	1
8	3^{+}_{-1-11}	$\begin{bmatrix}0&0&-1\\1&0&0\\0&-1&0\end{bmatrix}$	1
9	3^{-}_{111}	$\begin{bmatrix}0&1&0\\0&0&1\\1&0&0\end{bmatrix}$	1
10	3^{-}_{1-1-1}	$\begin{bmatrix}0&-1&0\\0&0&1\\-1&0&0\end{bmatrix}$	1
11	3^{-}_{-1-11}	$\begin{bmatrix}0&1&0\\0&0&-1\\-1&0&0\end{bmatrix}$	1

continued ...

Table 1

No.	tag	matrix (polar)	det
12	3^{-}_{-11-1}	$\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}$	1
13	m_{1-10}	$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	−1
14	m_{110}	$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	−1
15	-4^{+}_{001}	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	−1
16	-4^{-}_{001}	$\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}$	−1
17	m_{01-1}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	−1
18	-4^{+}_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$	−1
19	-4^{-}_{100}	$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$	−1
20	m_{011}	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{bmatrix}$	−1
21	m_{-101}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	−1
22	-4^{-}_{010}	$\begin{bmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	−1
23	m_{101}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$	−1
24	-4^{+}_{010}	$\begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$	−1